

Simulating the Future: The Role of Multimodal Virtual Reality in Futuristic Technologies' User Testing and Training

Hanna J. Neroj*

Georgia Institute of Technology

Keane Zhang†

Georgia Institute of Technology

Bruce N. Walker‡

Georgia Institute of Technology



Figure 1: A flying car flying over a simulated city with take-off and landing stations.

ABSTRACT

Immersive multimodal Virtual Reality (VR) simulations can effectively bring futuristic technologies to life, providing users with a meaningful insight into how the product will operate and feel, as well as support training for technologies that are still in development. This study investigates the effectiveness of VR experiences in shifting users' thinking from abstract to concrete, using flying cars as an illustrative example. The preliminary data suggest that multimodal VR facilitates a transition from abstract to

more concrete thinking, albeit potentially with adverse effects on individual risk perception. Areas of future research include larger-sample studies that will enable diversity in initial attitudes towards given technology, as well as diversity in personality characteristics.

Index terms: Virtual Reality, multimodal simulation, technology acceptance, risk perception.

1 INTRODUCTION

Rapid technological advancements present a challenge: how do we study user acceptance of, familiarization with, and training for novel products and services in the absence of the physical experience? When novel technologies (e.g., flying cars) lack mature prototypes to allow direct exposure, companies traditionally shift to techniques such as text descriptions, surveys, and interviews, and sometimes visual renderings. In terms of technology adoption, those tools capture one's feelings in the present. However, for complex, innovative transportation

* e-mail: hnero3@gatech.edu

† e-mail: keane@gatech.edu

‡ e-mail: bruce.walker@psych.gatech.edu

technologies, such feelings are formed based on a vague imagination of the technology, likely shaped by science fiction rather than one's own experience. This is likely to result in perceiving the technology as psychologically distant, i.e., that the technology will exist in the distant future, in a distant location, or only in a hypothetical scenario [1]. Therefore, surveys alone are insufficient to predict how one will feel once these technologies are developed and able to be experienced. And, of course, surveys, and even descriptions and renderings, are not able to help a potential user become familiar with, and trained on, futuristic technologies.

In contrast, immersive and interactive multimodal Virtual Reality (VR) simulations can effectively bring the futuristic technologies to life, in sufficient multisensory detail to give potential users a meaningful insight into how the product will operate and feel [2]. Being able to interact with and actually operate the technology (in VR), and have multisensory feedback, has the dual benefits of (a) supporting familiarity and training for the real technology (even before it exists!), and (b) reducing psychological distance, thereby causing individuals to assess the innovation on a more concrete and intellectual, rather than emotional, basis. On the other hand, there has been mixed success with VR in training when it comes to highly risky situations, such as timely response to natural disasters. For instance, a study by Simpson and colleagues [3] found adverse effects of using immersive VR on individual risk perception of flooding hazard, which the authors hypothesized could be associated with positive affective responses to technological immersion compared to real life.

In this paper, we use the flying car technology as an illustrative example of such intricate innovation within the transportation sector. We investigate the effectiveness of immersive interactive VR experiences, with visual, audio, and haptic/motion simulation, in shifting users' thinking from highly abstract to concrete. Additionally, we present qualitative and quantitative data from a small-sample study (N=4), addressing general beliefs about flying cars and the perceived risk associated with the use of flying cars.

2 BACKGROUND

Research on technology acceptance predominantly employs established technology acceptance models, such as the Technology Acceptance Model (TAM) [4] and the Unified Theory of Acceptance and Use of Technology (UTAUT) [5], along with their subsequent extensions. These models draw upon the user's experience with the technology and incorporate predictors such as perceived ease of use and perceived usefulness. However, this limitation, inherent in the existing models, presents a substantial challenge in the assessment of novel, innovative technologies that are currently in a purely conceptual stage.

2.1 Decision Making in the Futuristic Technology Context

2.1.1 Psychological Distance

Another challenge in assessing the acceptance of non-existent technologies is our evolving perception influenced by psychological distance. The Construal Level Theory (CLT) posits that when an event or object is perceived as distant (in time, location, socially, or hypothetically), people deploy high-level construals and engage in abstract thinking. Consequently, when the

object is psychologically proximal, people will deploy low-level construals and will engage in more concrete thinking, including more incidental features [1]. In our research, we employ multimodal virtual reality (VR) to bridge the gap between abstract and concrete thinking. We hypothesize that due to the inability to experience futuristic technologies (such as flying cars), individuals tend to conceptualize them abstractly. By utilizing VR simulation, participants will be able to experience given futuristic technologies "here and now," enabling concrete, low-level thinking.

2.1.2 Risk Perception

Perceived risk is defined as a subjective, intuitive judgment of harm influenced heavily by our emotions [6]. In this research, we argue that perceived risk will be especially important in the context of futuristic technologies' acceptance. Since futuristic technologies do not currently exist, there are no objective metrics that can quantify the potential for harm. Consequently, individuals may place greater emphasis on their subjective assessment of risk in guiding their decisions regarding whether to accept or reject a given technology. As VR gives a real-time sensory perception update through physical movement and the feeling of "being there" [7], it may trigger different emotional states and impact risk perception due to exposure to imminent threats inside the computer environment.

Research suggests that immersive VR may increase awareness of real-world dangers and replicate real-world risk-related behaviors better than when responding to a text-based survey [2]. Furthermore, there may be a difference in risk perception depending on the perceived psychological distance of futuristic technology pre- and post-exposure to a simulated VR environment.

3 SYSTEM DESIGN: THE FLYING CAR SIMULATOR

Our VR-based flying car simulator aims to minimize the psychological distance between the tester and the simulated car, thereby enhancing immersion and realism.

The hardware setup includes a Logitech G25 electronic steering wheel, pedals, an Xbox game controller, and an HTC VIVE VR Headset. For development and testing purposes, the simulation can also be run on a computer monitor and controlled with a keyboard. The simulation is built on Unity Engine [8], with various assets sourced from the Unity Store, including the flying car model, materials, and physics.

The operation of the car is divided into two modes: driving mode and flying mode. Care was taken to ensure driving mode closely mirrors the experience of driving a car in real life, while flying mode provides a clearly different control experience. The measures include calibrating the acceleration and braking to match the sensation of driving a car and having separate controls for driving and flying. Driving is controlled entirely with the steering wheel and pedals, whereas flying is controlled only with the game controller. Driving and flying mode employ different motion profiles, such as having different amounts of tilt while the car is accelerating in each mode. Multimodality is achieved through auditory cues and haptic feedback provided by the steering wheel and the pedals, reinforcing immersion.

The simulation is set in a virtual city populated with roads, buildings, and designated *Vertispots* for vertical takeoff and landing. The simulated environment supports navigation during both driving and flying modes in a familiar urban context.

To aid in ease of operation, various human-machine interface (HMI) elements are built into the car, including an altitude and

speed indicator, a text display indicating the operating mode of the car (e.g., driving, flying, takeoff, or landing), and a downward-facing camera that provides a view beneath the car during takeoff, flight, and landing.

Finally, to allow for an investigation into plausible future flying car use cases, such as autonomous air taxis, the car also allows for different levels of vehicular autonomy. This includes full manual control by the user, and self-driving mode with or without a simulated safety driver in the driver's seat.

4 METHOD & PRELIMINARY RESULTS

4.1 Design

To evaluate the preliminary efficacy of the VR flying car simulator, we have adopted a mixed-method approach. We have recruited N=4 participants (M=3, F=1, mean age=25.5 years) from a technical university on a voluntary basis. Each participant had a valid driver's license.

We first provided a brief overview of the flying car technology, followed by a short survey to assess participants' risk perception of flying cars. Afterwards, we conducted a 5-minute semi-structured interview about participants' beliefs and familiarity with flying cars. Participants were then exposed to four automation conditions in a random order: one that was full manual control by the participant, and three modes of autonomy as a passenger in the back seat: no autonomy (human-driven), fully autonomous with a safety driver who could take over control if necessary, and a fully autonomous flying car with no human driver present. Each of the three modes in which the participant acted as a passenger was pre-programmed with an identical route. In manual mode, participants were granted absolute autonomy in selecting their route. After the manipulation, participants returned to the survey where they answered the same questionnaire on risk perception, adapted from Walpole and Wilson [9]. Additionally, we collected data on the Locus of Control (LoC) using the Internal-External Locus of Control Short Scale-4 (IE4) [10] and Ten Item Personality Measure (TIPI) [11]. All interviews were conducted in a one-on-one setting.

4.2 Preliminary Results

4.2.1 Demographics

Table 1 below presents the demographic data along with the results of the IE-4 and the TIPI surveys.

Table 1: Summary of participant characteristics

Gender	Age	LoC	Dominant Personality Trait
Female	23	Internal	Conscientiousness
Male	29	Internal/Balanced	Agreeableness
Male	22	Internal	Agreeableness
Male	28	Internal	Agreeableness

No differences were observed among participants in their psychological profiles (LoC, dominant personality trait). Consequently, these findings serve as a potential limitation of the collected data.

4.2.2 Pre- vs. Post-Exposure Attitudes Towards Flying Cars

All participants expressed moderate to high excitement about flying cars in the pre-exposure interview. In particular, all participants exhibited positive beliefs and trusting attitudes towards major technology companies, suggesting that they would trust the safety of their products. Two of four participants have used a fully autonomous vehicle before (i.e., a self-driving Tesla).

Figure 2 depicts the mean perceived risk scores pre-exposure (on the left) and post-exposure (on the right). Each line indicates one participant. For all participants, perceived risk scores followed a negative linear trend from pre- to post-exposure. In the post-exposure interview, participants exhibited heightened anxiety levels in the fully autonomous scenario, in contrast to human-driven and safety driver modes. This was consistently observed across all participants, precluding any potential order effect biases.

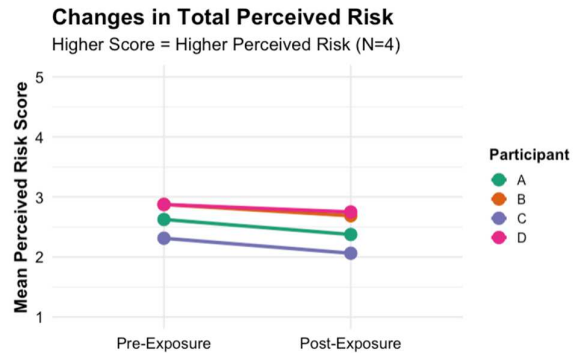


Figure 2: Perceived risk scores pre-exposure (left) and post-exposure (right).

One participant described a "strange sense of comfort" when the driver was present compared to fully autonomous mode. Another participant felt the need to be more alert in autonomous mode due to past experience with their fully autonomous car failing.

4.2.3 Simulator Realism

In addition to participants' attitudes towards flying cars, we collected qualitative feedback on the design of the simulator. All participants perceived the duration of each simulated scenario (3 minutes) to be suitable for an immersive experience and affirmed that it was adequate to concentrate on the change in their emotional state with each mode change. Participants suggested that the scaled-down design of the city negatively impacted the perceived realism. In the flying mode, the primary view was of the horizon rather than the buildings. Furthermore, the controls for the flying mode induced participants to perceive the driving mode as a game rather than an authentic driving experience. Consequently, all participants exhibited a preference for scenarios where they acted as a passenger, as they felt a stronger connection to the "real-life experience of a taxi ride." Finally, participants expressed doubt that controlling the flying car will be "as easy as one button".

5 DISCUSSION

The preliminary results suggest the immersive VR experience allows for a shift in user thinking about a technology from abstract to more concrete. In particular, participants noted that after the simulation, more salient examples of "things that could go wrong" emerged. Moreover, participants seemed to think of more practical

concerns, e.g., "How will flying cars integrate with traffic?" "How will the car actually behave mid-air?" "I don't think switching to flight mode will actually be just one button." Interestingly, despite the concerns that were communicated during the interviews, the risk perception from pre-exposure to post-exposure followed a negative linear trend. Different autonomy modes influenced users' emotional states. All participants reported a heightened need for alertness in autonomous modes, either due to their experience with self-driving cars or heightened anxiety in fully autonomous modes when the driver was absent. Notably, participants expressed a lack of logical basis for anxiety, as it did not align with their positive attitude towards flying cars prior to exposure. Post-exposure, participants verbally indicated their willingness to utilize flying cars should they become available, confirming their initial attitudes, despite the emotional change during the simulation.

Basing our rationale on the Construal Level Theory [1], we hypothesized that participants would perceive flying cars as less risky in the pre-exposure condition (abstract thinking), compared to the post-exposure condition (concrete thinking). Even though participants indicated the shift from abstract to more concrete thinking, mentioning more thinking of practical use cases and specific concerns, contrary to our hypothesis, the post-exposure perceived risk scores were lower. There may be several reasons for this observation.

A plausible explanation might be that the majority of our participants conformed to the dominant personality type, and all of them demonstrated an internal LoC. Both the agreeableness and conscientiousness traits and internal LoC have been found in prior research as indicators of higher technology acceptance. For instance, Tseng and Hsia found that internal LoC has a significant positive impact on technology acceptance [12]. Calluso and Devetag [13] found that conscientious people may be more willing to engage with novel technology (Artificial Intelligence), as they see such technologies as a means of improving reliability and efficiency. Another study found agreeableness to be positively associated with the use of advanced technologies for socializing [14]. These findings seem consistent with our data, where participants expressed a trusting attitude toward advanced technologies as well as companies working on them. One participant expressed that they would trust an autonomous flying car with a safety driver more than they would trust a human driver with no support. However, to confirm this hypothesis, a larger-N study should be conducted, with a recruitment of a demographically diverse sample that would allow for assessing the differences in the risk perception based on varying LoC and different personality characteristics. Furthermore, all participants exhibited a positive attitude towards flying cars pre-exposure, and the VR simulation appeared to reinforce these attitudes, potentially explaining the adverse trend in risk perception. A more diverse sample would elucidate how these perceptions evolve in response to initial attitudes that are more neutral or negative.

6 CONCLUSION

Although preliminary, these findings suggest several significant advantages of employing multimodal VR technologies for training purposes. Multimodal VR appears to shift users' thinking from abstract to more concrete thinking. It enables participants to test more practical examples and, most importantly, it facilitates the experience of diverse emotions "here and now," rather than acting based on preferences that may differ from actual experiences.

It is noteworthy that in our study, participants initially exhibited enthusiasm for flying cars. The exposure through immersive VR appeared to reinforce these preferences, as indicated by decreased

risk perception scores. However, a lowered perceived risk is a similar effect as was observed in the Simpon et al. study [3], where immersive VR had an adverse impact on risk perception of a hazardous event. In the context of complex transportation technologies that harbor high potential for adverse outcomes, a decreased perceived risk may not be a desirable outcome. Consequently, further research should encompass hazardous scenarios that can enhance training and reinforce corrective safety behaviors despite the potential adverse effects on individual perceived risk that immersive VR may entail.

Finally, further research is required to assess how risk perception varies depending on individual personality characteristics and whether positive or negative pre-exposure attitudes toward technology influence risk perception.

FIGURE CREDITS

Figure 1 screen capture of Unity, ownership of GT Sonification Lab.

Figure 2 screen capture from R Studio, produced by the authors.

ACKNOWLEDGMENTS

The authors wish to thank numerous Georgia Institute of Technology students in the Sonification Lab for their invaluable contributions to the development of the virtual reality (VR) environment and the creation of diverse Unity assets. A particular acknowledgment is extended to Keane Zhang for the most recent and substantial contributions to the design of the flying car asset.

REFERENCES

- [1] Trope, Y., & Liberman, N. (2010). Construal-level theory of psychological distance. *Psychological review*, 117(2), 440. doi: 10.1109/TVCG.2016.2615308
- [2] Slater, M., & Sanchez-Vives, M. V. (2016). Enhancing our lives with immersive virtual reality. *Frontiers in Robotics and AI*, 3, 74. doi: 10.3389/frobt.2016.00074
- [3] Simpson, M., Padilla, L., Keller, K., & Klippel, A. (2022). Immersive storm surge flooding: Scale and risk perception in virtual reality. *Journal of Environmental Psychology*, 80, 101764.
- [4] Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly*, 319-340.
- [5] Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS quarterly*, 425-478.
- [6] Slovic, P., Finucane, M. L., Peters, E., & MacGregor, D. G. (2007). The affect heuristic. *European journal of operational research*, 177(3), 1333-1352. <https://doi.org/10.1016/j.ejor.2005.04.006>
- [7] Sanchez-Vives, M. V., & Slater, M. (2005). From presence to consciousness through virtual reality. *Nature reviews neuroscience*, 6(4), 332-339.
- [8] Unity Technologies, *Unity*. (2024). [Online]. Available: <https://unity.com/>
- [9] Walpole, H. D., & Wilson, R. S. (2021). A yardstick for danger: Developing a flexible and sensitive measure of risk perception. *Risk analysis*, 41(11), 2031-2045. <https://doi.org/10.1111/risa.13704>

- [10] Nießen, D., Schmidt, I., Groskurth, K., Rammstedt, B., & Lechner, C. M. (2022). The Internal–External Locus of Control Short Scale–4 (IE-4): a comprehensive validation of the English-language adaptation. *PLoS One*, 17(7), e0271289. <https://doi.org/10.1371/journal.pone.0271289>
- [11] Gosling, S. D., Rentfrow, P. J., and Swann, W. B. (2003). A very brief measure of the Big-Five personality domains. *Journal of Research in personality*, 37(6), 504–528. doi: 10.1016/s0092-6566(03)00046-1
- [12] Tseng, A. H., & Hsia, J. W. (2008, September). The impact of internal locus of control on perceived usefulness and perceived ease of use in e-learning: An extension of the technology acceptance model. In *2008 International Conference on Cyberworlds* (pp. 815-819). IEEE. doi: 10.1109/CW.2008.109
- [13] Calluso, C., & Devetag, M. G. (2025). The impact of technology acceptance and personality traits on the willingness to use AI-assisted hiring practices. *International Journal of Organizational Analysis*, 33(5), 1368-1385. doi: 10.1108/IJOA-06-2024-4562
- [14] G, S., Chakraborty, D., Polisetty, A., Khorana, S., & Buhalis, D. (2023). Use of metaverse in socializing: Application of the big five personality traits framework. *Psychology & Marketing*, 1–19. <https://doi.org/10.1002/mar.21863>